Safal Kumar Shrestha

AI/ML Engineer | Computer Vision, RAG and Model Integration

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Professional Summary

AI undergraduate developing PyTorch computer-vision features, LangGraph/RAG agents, and Python services that connect models to applications. Contributed to 3 shipped apps and operate 5 self-hosted workloads alongside full-time study. Experience spans data preparation, model integration, testing, and GPU infrastructure investigation, with STEAM/DFRobot-supported sensor-control research.

Education

B.Tech in Artificial Intelligence 2022 – Present
Kathmandu University, Nepal | Full-time | Expected 2027 | CGPA: 3.81
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Large Language Models, Probability and Statistics, Data Structures and Algorithms.

Diploma in Computer Engineering Completed 2022
Nepal Polytechnic Institute, Nepal | 81.62%

Technical Skills

Programming: Python, SQL, C/C++, Java, JavaScript
Machine Learning: PyTorch, TensorFlow, scikit-learn, model training, NLP, graph neural networks
Data and Evaluation: Preprocessing, model evaluation, error analysis, test cases, data pipelines
Computer Vision: OpenCV, YOLOv8, Tiny U-Net, image processing, detection, segmentation
Generative AI: LangGraph, LLMs, RAG, Qdrant, embeddings, prompt engineering, tool calling
API Familiarity: OpenAI, Anthropic, Gemini; structured outputs and tool calling
Backend and Data: FastAPI, Django/DRF, REST APIs, MySQL, MongoDB, Redis, RabbitMQ
ML Tools / Deployment: MLflow (basics), Docker, Kubernetes, Linux, Git, CI/CD, monitoring
IoT and Integration: ESP32, Arduino, PlatformIO, MQTT, WebSockets, data pipelines

Development and Research Experience

Independent AI/ML and Application Developer **2022 – Present**
Google Play publisher name: bala.ai | Part-time

- Contributed services, data preparation, model integration, tests, and releases for 3 shipped apps: KU Krishi, TrekSathi, and Tiffin, alongside full-time study.
- Build Python services and data pipelines connecting databases, REST APIs, and model inference; investigate incorrect outputs and integration failures using test inputs, logs, and feedback.
- Operate 5 self-hosted workloads; integrate databases and secure endpoints with SSL/TLS, reverse proxies, and firewalls.

ML, Computer Vision and IoT Researcher **2026**
Hydroponics AI, Kathmandu University | Part-time

- Secured STEAM research funding and DFRobot hardware sponsorship for a hydroponics automation prototype.
- Built 6 sensing/control workflows across 2 ESP32 platforms for monitoring, pumping, and nutrient dosing; integrated pH, electrical conductivity, and carbon dioxide readings via MQTT.
- Built a monitoring/control app connecting sensor data, actuator control, and manual overrides through FastAPI, MQTT, and WebSockets; tested behavior and documented findings.

High-Performance Computing Core Engineer **2025**
AI HPC Initiative, Kathmandu University | Part-time

- Configured Kubernetes and HAProxy to route workloads and support distributed AI workers.
- Investigated RTX 3080 Ti topology, allocation, and virtualization limits; documented hardware, team-knowledge, and physical-infrastructure constraints to guide deployment decisions.

Selected AI/ML and Infrastructure Projects

TrekSathi / TrekMate — Travel Platform and AI Agent **Jul 2025 – Present**

- Built TrekMate with LangGraph, RAG, and Qdrant, combining user input and trail context with geospatial/weather tools for preference-based itinerary planning.
- Integrated Django/MongoDB services for 650+ destinations and 3,000+ accommodations; the surrounding Flutter application reached 150+ users.

KU Krishi — Agricultural AI Assistant

- Built PyTorch Leaf Color Chart, late-blight prediction, and blight segmentation features; connected preprocessing and inference to FastAPI/Django services and a Flutter app.
- Integrated a LangGraph agent with crop-disease tools and market information; the app reached 180+ Google Play downloads and was a National ICT Award semi-finalist in 2024.

Satellite-Image Land Segmentation **Jun 2026 – Jul 2026**

- Preprocessed satellite imagery with Python and compared segmentation outputs for GIS analysis and model experimentation.

Wilms Tumor Segmentation **Jun 2025 – Jul 2025**

- Developed a Tiny U-Net for medical-image tumor segmentation in an academic Pattern Recognition project.

Self-Hosted Application Platform — Linux, Docker, MinIO

- Provisioned bare-metal application hosting and S3-compatible storage; secured endpoints using reverse proxies, SSL/TLS, and firewalls.

Ongoing Research

Autonomous Driving with CARLA Aug 2026 – Present
Explore reinforcement learning for vehicle decision-making and control in simulation, with an emphasis on safe, efficient, and traffic-compliant behavior.

Reinforcement Learning for Software-Defined Networking Aug 2026 – Present
Explore reinforcement learning for intelligent network control, optimization, and decision-making.

Leadership, Teaching and Mentoring

Operations Lead, AI Convergence, Kathmandu University 2025 – Present
Coordinate teams, logistics, schedules, and technical requirements; supported National AI Summit (Aug 2025) and AI Conclave (Jan 2026).

Mentor, DataCamp / AI CANVAS, KUAIC Dec 21–28, 2025
Instructor, Python X Git Bootcamp, KUAIC Jul 10–13, 2025

Certifications, Awards and Funding

LangGraph Certificate, LangChain Academy.
STEAM Research Grant and DFRobot Hardware Sponsorship, Hydroponics AI (2026).
AI Crusade, Third Place (2024); Highest Score, KUAIC Python Bootcamp Coding Test (2023).
PCB Design and Embedded Systems Workshop, Kathmandu University (Apr 2026).